

MySQL – database

Data: collection of information or message

Data - Collection of raw files

Data does not give any meaning

For example, data – python Karthikeya 95

Information - arranging the data in structure format

Where information gives meaning

Information – course name marks

Database: storing data in a structured format and managing data in an electronic system(laptops)

Paper-based or book-based

Data security

Data backup - no recovery

Data redundancy – duplicate

Data transformation – from one place to another place

Data access

Data files – stores in system

Data security is somewhat high

Data backup - we can do it from the recycle bin

Data redundancy – duplicate

Data transformation

Data access - one by one

Data bases

Data security – high security

Data backup

Data redundancy – not possible

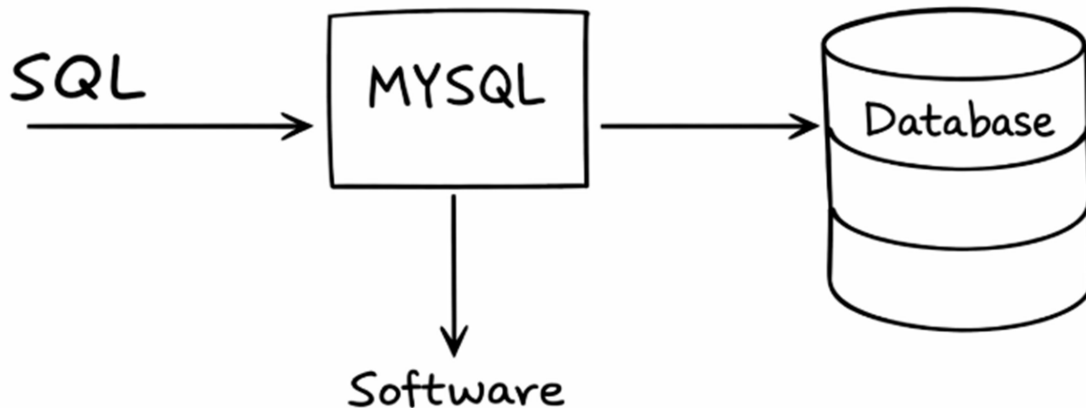
Data transformation – multiple persons can access

Data access – SQL – speed

SQL - structured query language

MySQL - software - DBMS data base management system

Postgres SQL



DBMS – DATA BASE MANAGEMENT SYSTEM - it is a software to manipulate data bases

Data Manipulate – data access, update, remove, storing

Data base models:

Relational Database Model (RDBMS)

NoSQL Database Model (Non-Relational)

Hierarchical Database Model

Network Database Model

Cloud Database

Object-Oriented Database Model (OODBMS)

Distributed data base model

Time Series Database (TSDB)

RDBMS:

Storing data in terms of table

- **RDBMS** → Type of database system (table format)
- **SQL** → Language to query the database
- **MySQL** → RDBMS software that understands SQL

ID	Name	Salary	Age
101	Ravi	45	22
102	Suresh	60	25
103	Balun	70	30

Column – fields

Rows – records

Storing structured data

structured data – tabular format

Unstructured data – not in tabular format: emails, msg

Semi structured data – Json file in key value pairs

NOSQL or NO – RDBMS:

Unstructured data – videos, images, audio

It stores unstructured data

MongoDB software to handle NoSQL

Eg: Instagram

User data – **RDBMS**

Post data – NoSQL

Hierarchical - level by level

Network base – graph

Object oriented data base management – objects

Distributed data base model - Data is stored in **multiple locations (multiple computers / servers)**

Time Series Database (TSDB) -

Timestamp Value Location

2026-02-28 09:00 25°C Hyderabad

Cloud Database: It is hosted and managed by cloud providers

SQL commands / sub language

SQL

DDL – **Data Definition Language**

DDL → Defines structure – CREATE, ALTER, DROP, TRUNCATE, RENAME

CREATE – Create database/table

ALTER – Modify table structure

DROP – Delete table/database

TRUNCATE – Remove all records (keeps structure)

RENAME – Rename table

DML – **Data Manipulation Language**

DML → Modifies data – INSERT, UPDATE, DELETE

INSERT → Add new records

UPDATE → Modify existing records

DELETE → Remove records

DQL – **Data Query Language**

DQL → Retrieves data – SELECT

SELECT * FROM students;

DCL – **Data Control Language**

DCL → Controls access

TCL – **Transaction Control Language**

TCL → Manages transactions

